

Sheth L.U.J & Sir M.V College Of Science

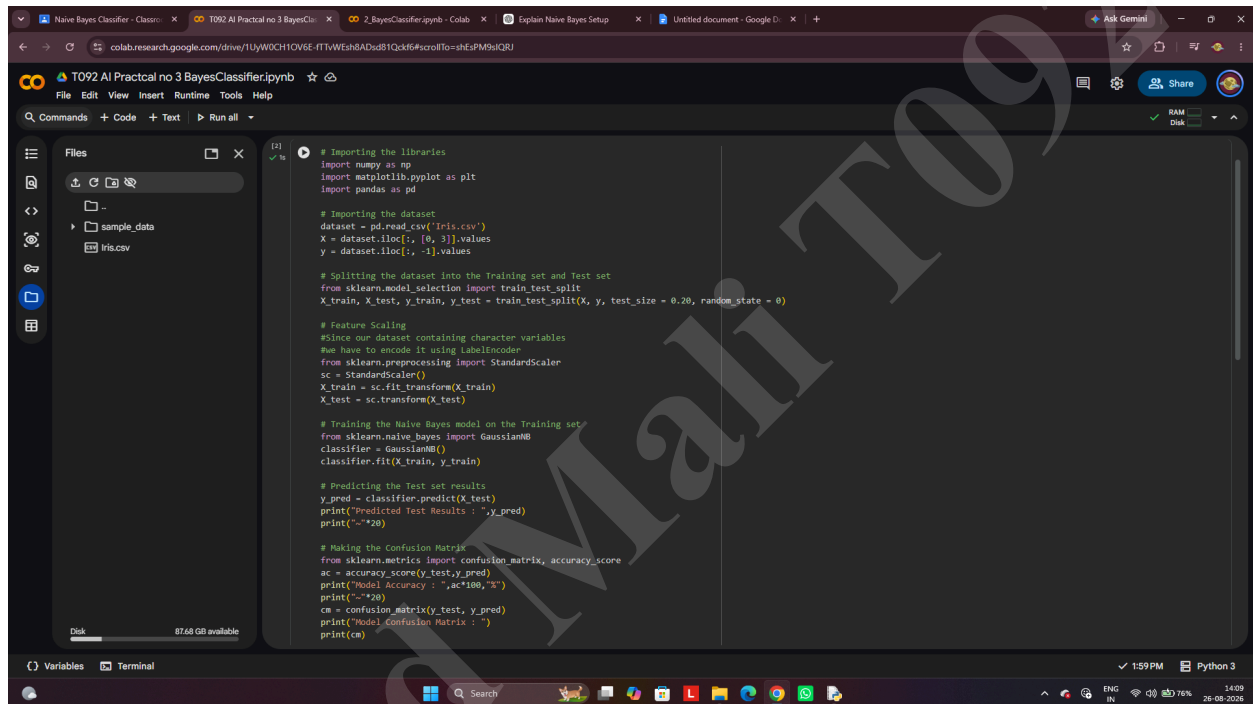
Subject :- Artificial Intelligences

Practical no 3

Aim :- Naive Bayes' Classifier

- Implement the Naive Bayes' algorithm for classification.
- Train a Naive Bayes' model using a given dataset and calculate class probabilities.
- Evaluate the accuracy of the model on test data and analyze the results.

Code :-



```
[1] In: # Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Importing the dataset
dataset = pd.read_csv('Iris.csv')
X = dataset.iloc[:, [0, 3]].values
y = dataset.iloc[:, -1].values

# Splitting the dataset into the training set and test set
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.20, random_state = 0)

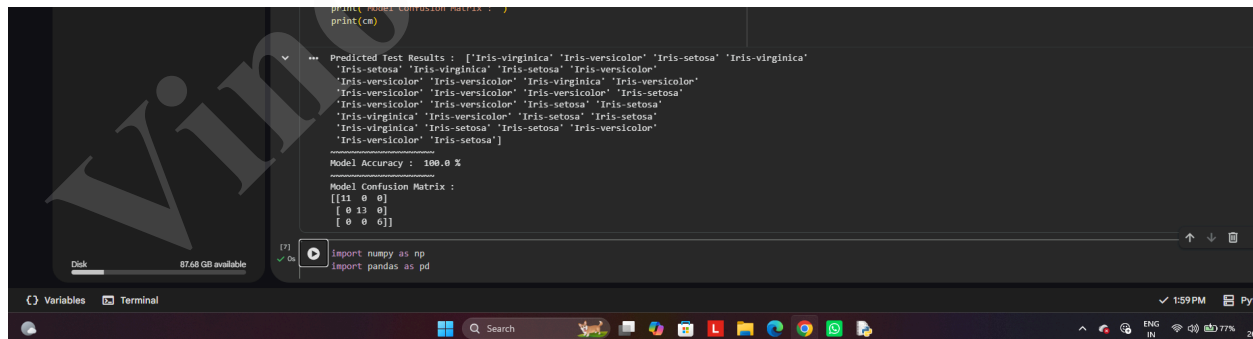
# Feature Scaling
# Since our dataset containing character variables
# we have to encode it using LabelEncoder
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)

# Training the Naive Bayes model on the Training set
from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)

# Predicting the Test set results
y_pred = classifier.predict(X_test)
print("Predicted Test Results : ", y_pred)
print("\n")

# Making the Confusion Matrix
from sklearn.metrics import confusion_matrix, accuracy_score
ac = accuracy_score(y_test, y_pred)
print("Model Accuracy : ", ac*100, "%")
print("\n")
cm = confusion_matrix(y_test, y_pred)
print("Model Confusion Matrix : ")
print(cm)
```

OutPut :-



```
[2] Out: Predicted Test Results : ['Iris-virginica' 'Iris-versicolor' 'Iris-setosa' 'Iris-virginica'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-versicolor'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa'
'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa' 'Iris-setosa'
'Iris-virginica' 'Iris-versicolor' 'Iris-setosa' 'Iris-setosa'
'Iris-virginica' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa']

Model Accuracy : 100.0 %

Model Confusion Matrix :
[[11  0]
 [ 0 13]
 [ 0  0]]

[3] In: import numpy as np
import pandas as pd
```

Name :- Vinod Manohar Mali
Roll No :- T092

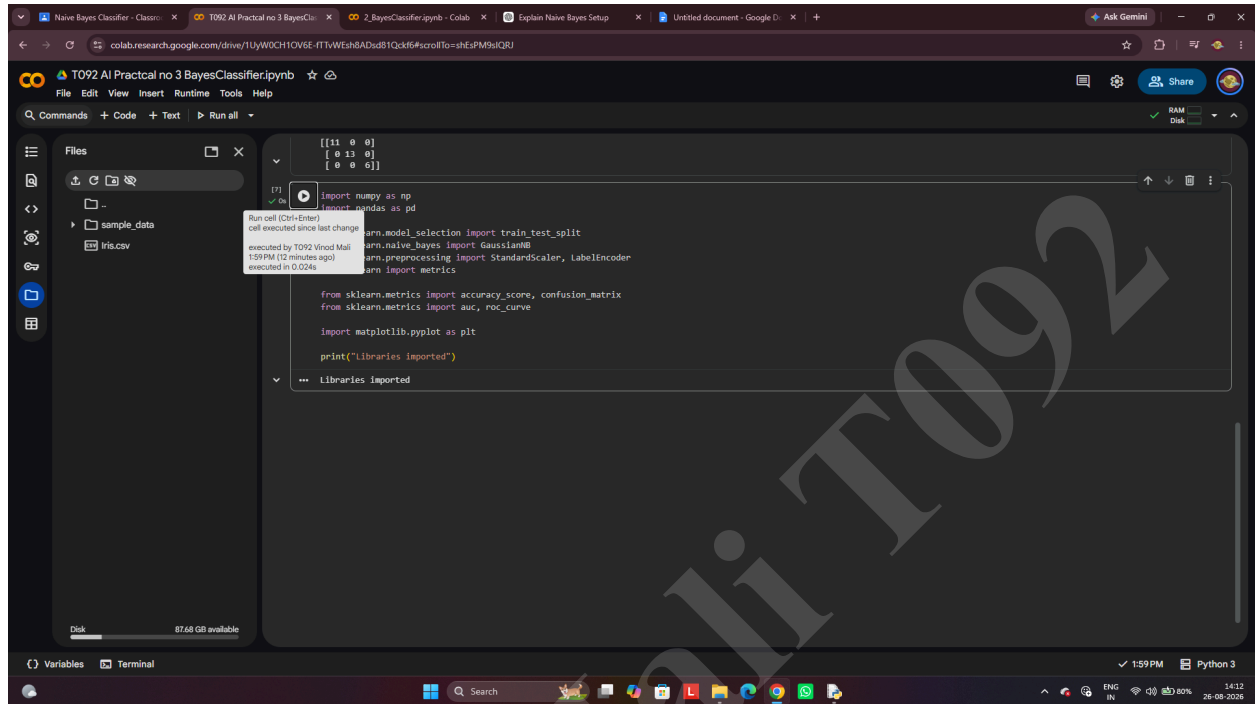
Sheth L.U.J & Sir M.V College Of Science

Subject :- Artificial Intelligences

Practical no 3

Code & OutPut :-

1. Import Libraries



The screenshot shows a Jupyter Notebook interface with a file explorer on the left displaying 'sample_data' and 'Iris.csv'. The code cell contains the following Python code:

```
[[11 0 0]
 [ 0 13 0]
 [ 0 0 6]]

import numpy as np
import pandas as pd

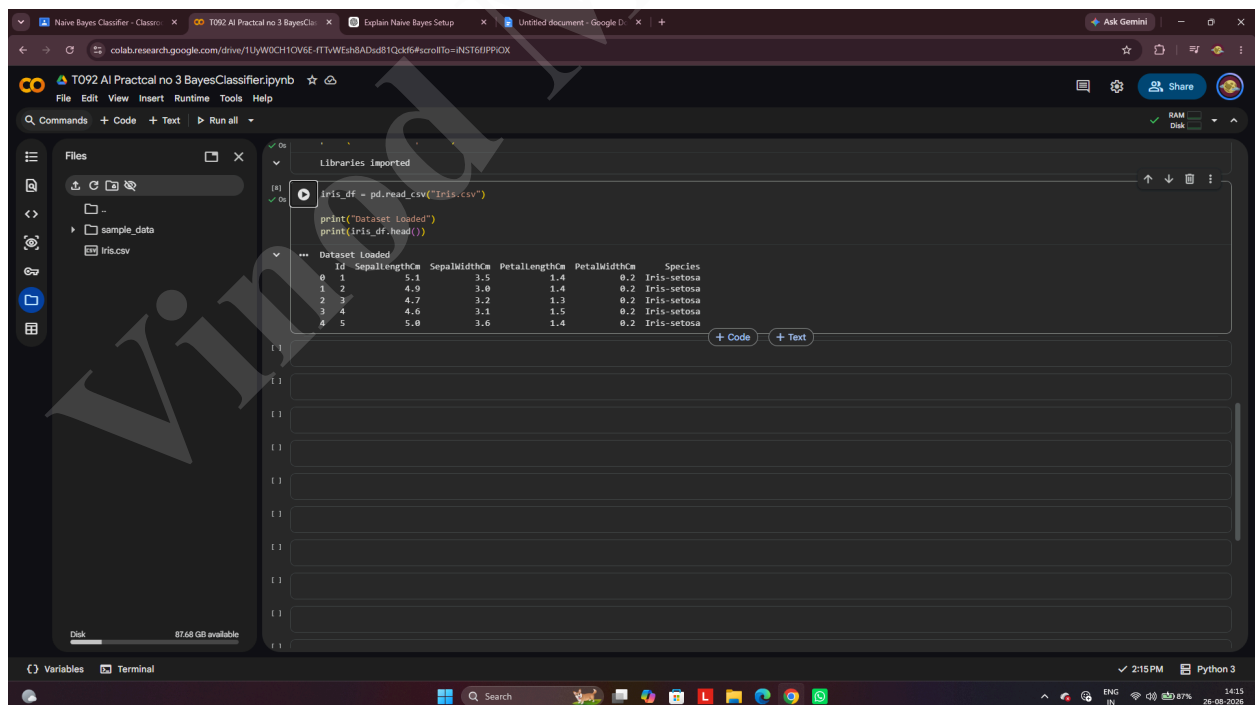
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.metrics import auc, roc_curve

import matplotlib.pyplot as plt

print("Libraries imported")
```

The output of the code cell shows the result of the imports, with a message 'Libraries imported' printed at the bottom.

2. Load the CSV Dataset



The screenshot shows the same Jupyter Notebook interface. The code cell now contains the following Python code:

```
Libraries imported

iris_df = pd.read_csv("Iris.csv")

print("Dataset Loaded")
print(iris_df.head())
```

The output of the code cell shows the result of loading the dataset, with a message 'Dataset Loaded' printed and the first five rows of the dataset displayed in a table:

	Id	SepallengthCm	SepalwidthCm	PetallengthCm	PetalwidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Name :- Vinod Manohar Mali

Roll No :- T092

3. Separate Input and Output

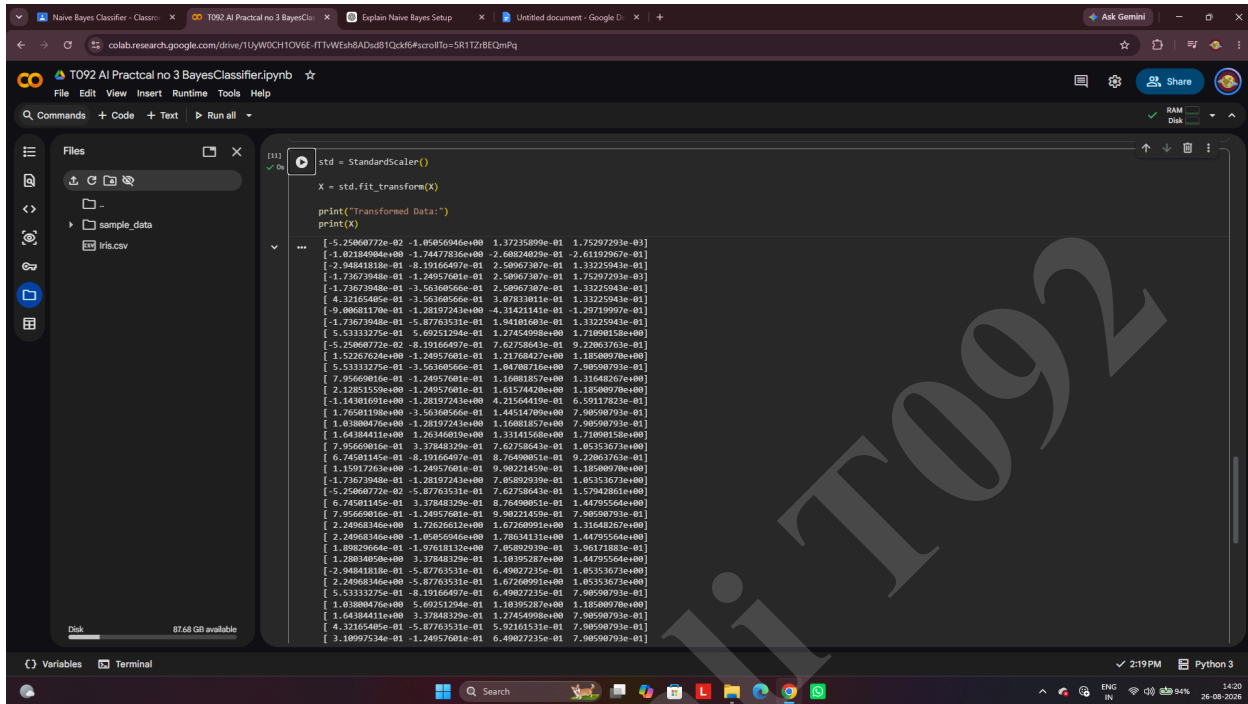


Sheth L.U.J & Sir M.V College Of Science

Subject :- Artificial Intelligences

Practical no 3

5. Standardize the Data

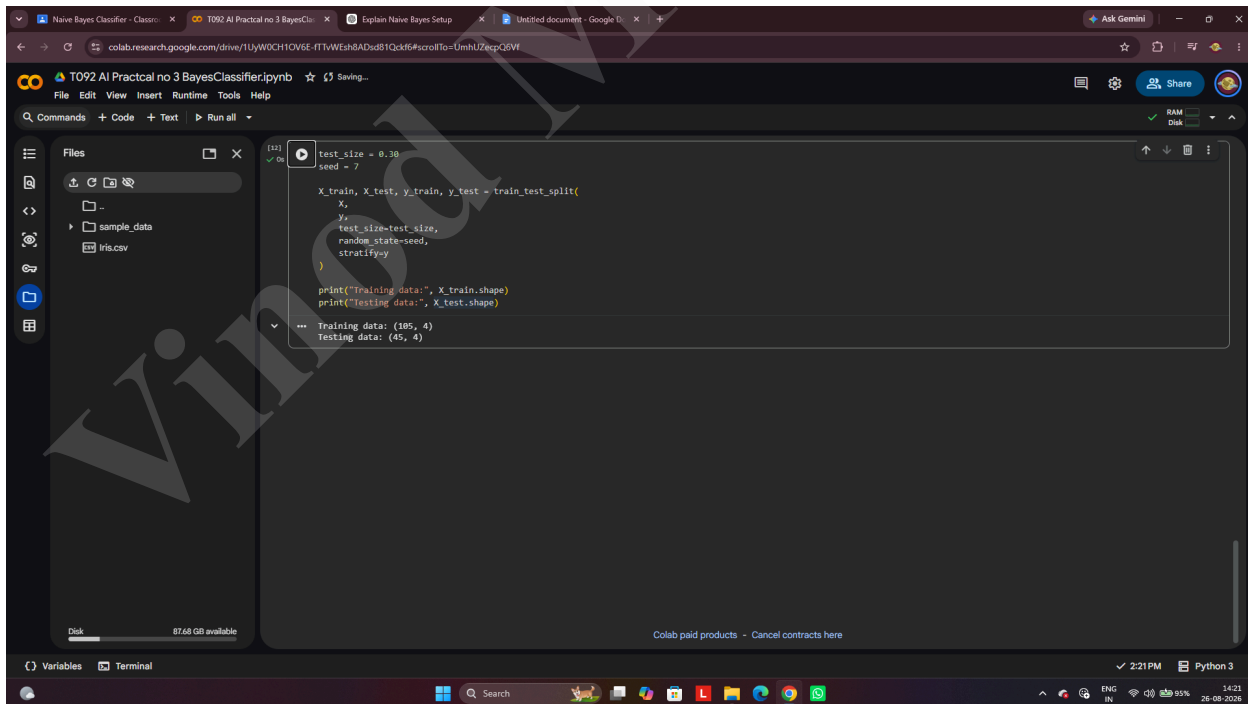


The screenshot shows a Google Colab notebook titled "T092 AI Practical no 3 BayesClassifier.ipynb". The code in the cell is as follows:

```
[11]: std = StandardScaler()
      X = std.fit_transform(X)
      print("Transformed Data:")
      print(X)
```

The output of the code is a large array of standardized data values, such as [-5.25060772e-02, -1.05050946e+00, 1.37235899e-01, 1.75207293e-03]. The notebook interface includes a file explorer on the left showing "sample_data" and "iris.csv", and a terminal at the bottom.

6. Split Training and Testing Data



The screenshot shows a Google Colab notebook titled "T092 AI Practical no 3 BayesClassifier.ipynb". The code in the cell is as follows:

```
[12]: test_size = 0.30
      seed = 7

      X_train, X_test, y_train, y_test = train_test_split(
          X,
          y,
          test_size=test_size,
          random_state=seed,
          stratify=y
      )

      print("Training data:", X_train.shape)
      print("Testing data:", X_test.shape)
```

The output of the code is:

```
Training data: (105, 4)
Testing data: (45, 4)
```

The notebook interface includes a file explorer on the left showing "sample_data" and "iris.csv", and a terminal at the bottom.

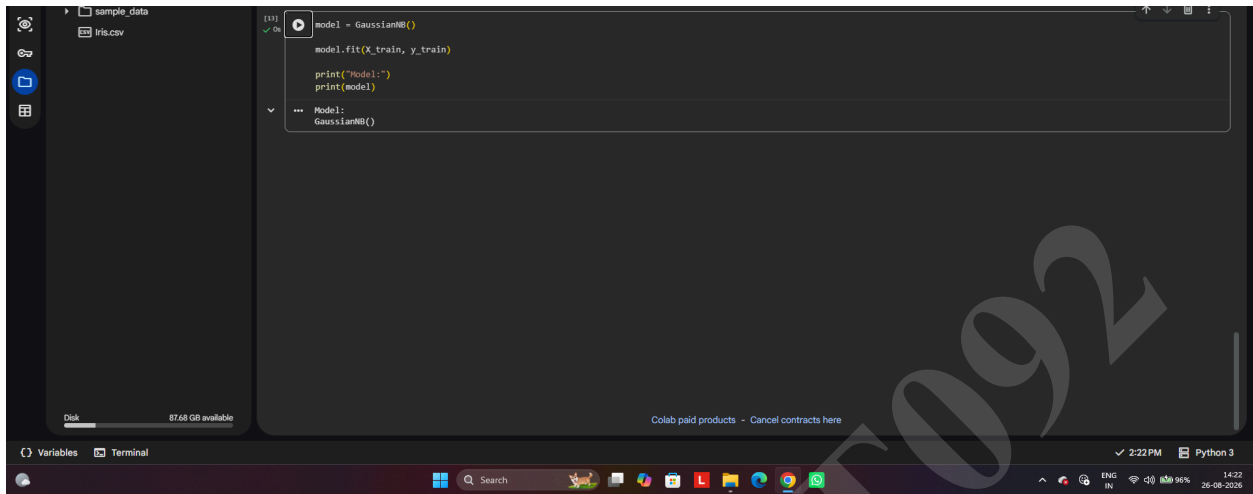
Name :- Vinod Manohar Mali
Roll No :- T092

Sheth L.U.J & Sir M.V College Of Science

Subject :- Artificial Intelligences

Practical no 3

7. Create and Train Gaussian Naive Bayes Model



The screenshot shows a Jupyter Notebook interface. On the left, a file explorer shows a folder named 'sample_data' containing a file 'iris.csv'. The main notebook area contains the following Python code:

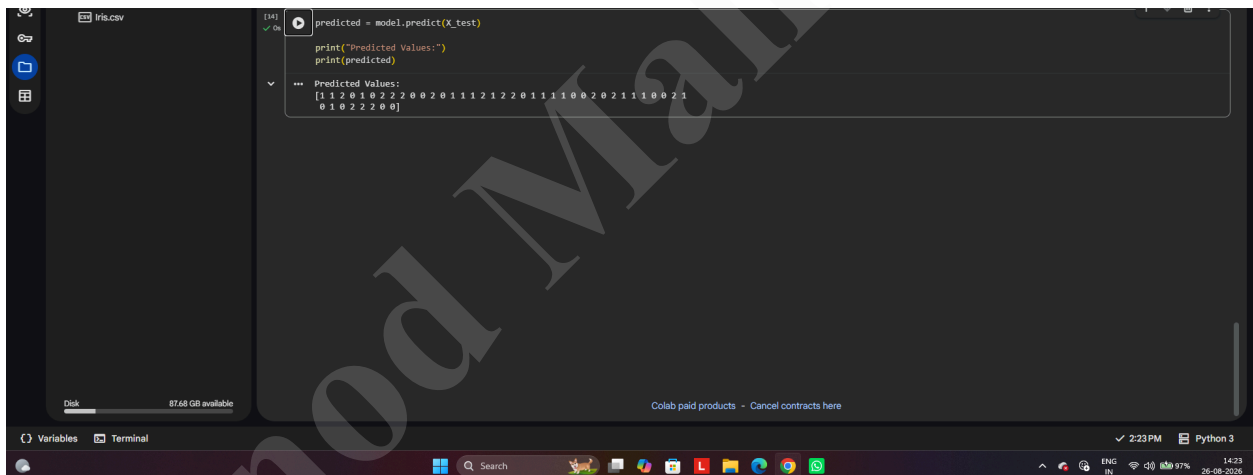
```
[13]: model = GaussianNB()
      model.fit(X_train, y_train)
      print("Model:-")
      print(model)
```

The output of the code is displayed below the cell:

```
Model:-
GaussianNB()
```

The bottom status bar indicates '2:22 PM', 'Python 3', and '14:22 26-08-2026'.

8. Make Predictions



The screenshot shows the same Jupyter Notebook interface. The code cell now contains:

```
[14]: predicted = model.predict(X_test)
      print("Predicted Values:")
      print(predicted)
```

The output shows the predicted values for the test set:

```
Predicted Values:
[1 1 2 0 1 0 2 2 0 0 2 0 1 1 1 2 1 2 2 0 1 1 1 1 0 0 2 0 2 1 1 1 0 0 2 1
 0 1 0 2 2 0 0]
```

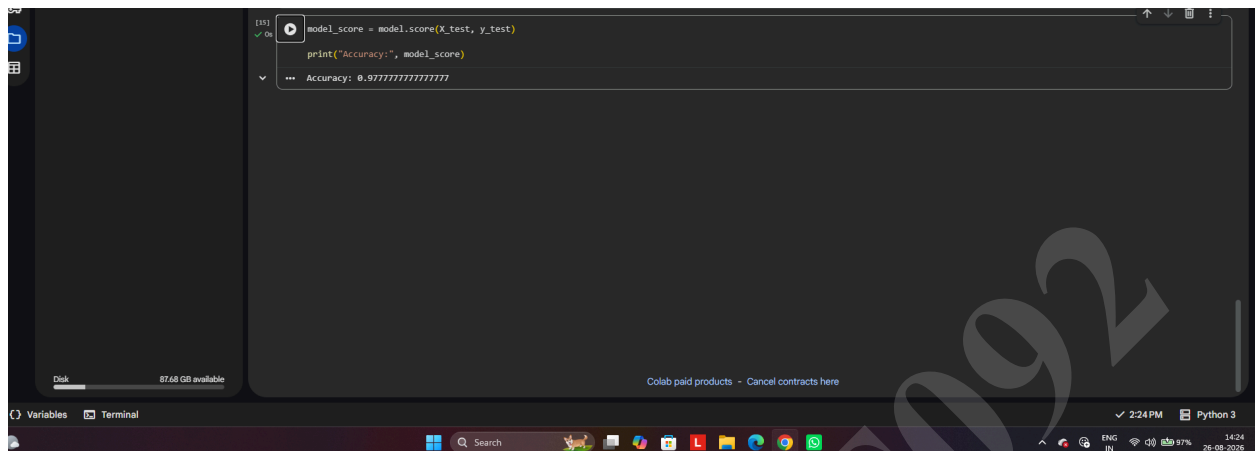
The bottom status bar indicates '2:23 PM', 'Python 3', and '14:23 26-08-2026'.

Name :- Vinod Manohar Mali

Roll No :- T092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Artificial Intelligences
Practical no 3

9. Calculate Accuracy



The screenshot shows a Jupyter Notebook interface with a dark theme. A code cell is selected, displaying the following Python code:

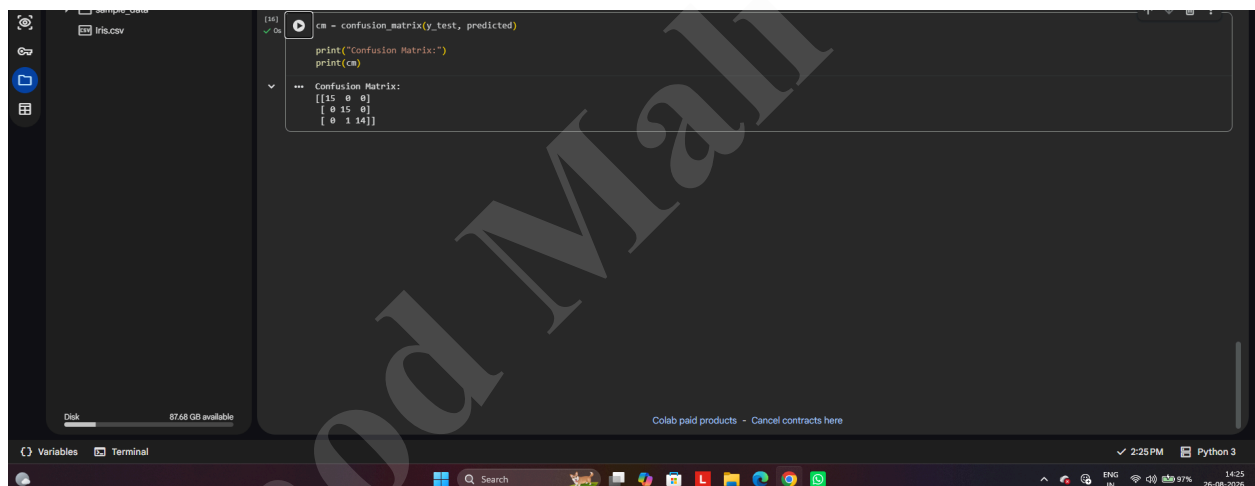
```
[15]: model_score = model.score(X_test, y_test)
      print("Accuracy:", model_score)
```

The output of the cell is displayed below the code:

```
Accuracy: 0.9777777777777777
```

The bottom status bar of the notebook shows "2:24 PM" and "Python 3".

10. Confusion Matrix



The screenshot shows a Jupyter Notebook interface with a dark theme. A code cell is selected, displaying the following Python code:

```
[16]: cm = confusion_matrix(y_test, predicted)
      print("Confusion Matrix:")
      print(cm)
```

The output of the cell is displayed below the code:

```
Confusion Matrix:
[[15  0  0]
 [ 0 15  0]
 [ 0  1 14]]
```

The bottom status bar of the notebook shows "2:25 PM" and "Python 3".

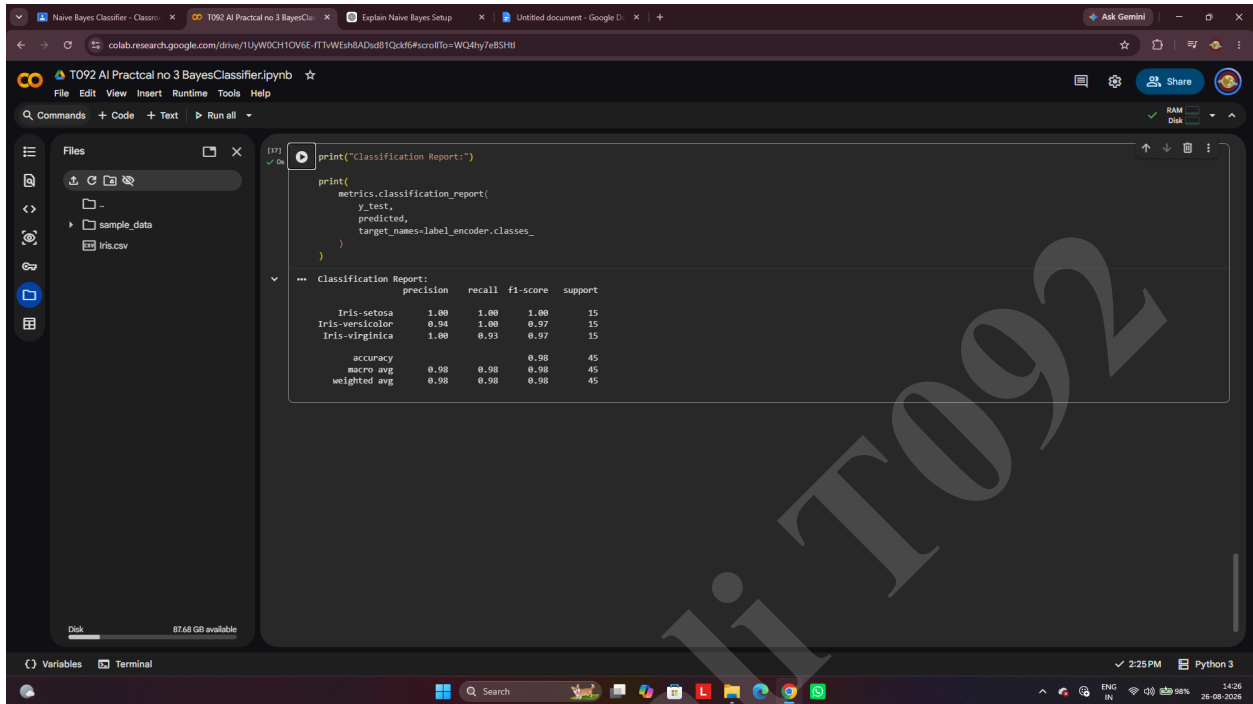
Name :- Vinod Manohar Mali
Roll No :- T092

Sheth L.U.J & Sir M.V College Of Science

Subject :- Artificial Intelligences

Practical no 3

11. Classification Report

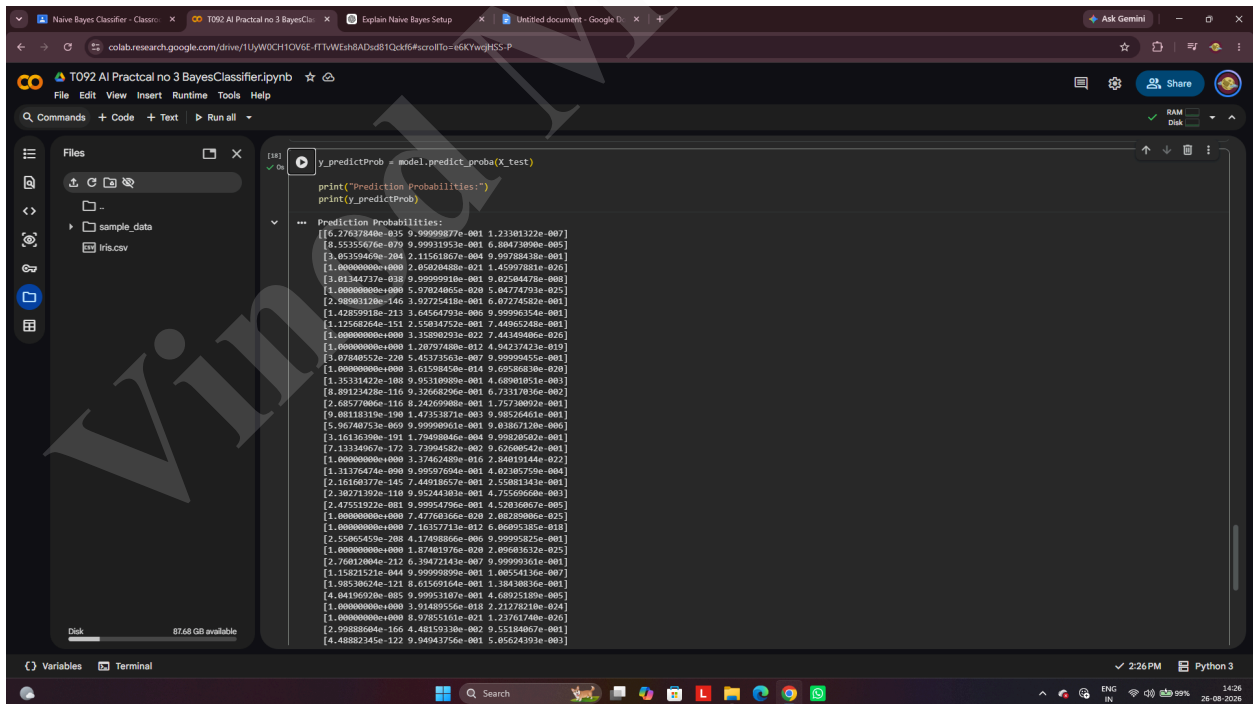


```
print("Classification Report:")

print(
    metrics.classification_report(
        y_test,
        predicted,
        target_names=label_encoder.classes_
    )
)
```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	15
Iris-versicolor	0.94	1.00	0.97	15
Iris-virginica	1.00	0.93	0.97	15
accuracy			0.98	45
macro avg	0.98	0.98	0.98	45
weighted avg	0.98	0.98	0.98	45

12. Prediction Probabilities



```
y_predictProb = model.predict_proba(X_test)

print("Prediction Probabilities:")
print(y_predictProb)
```

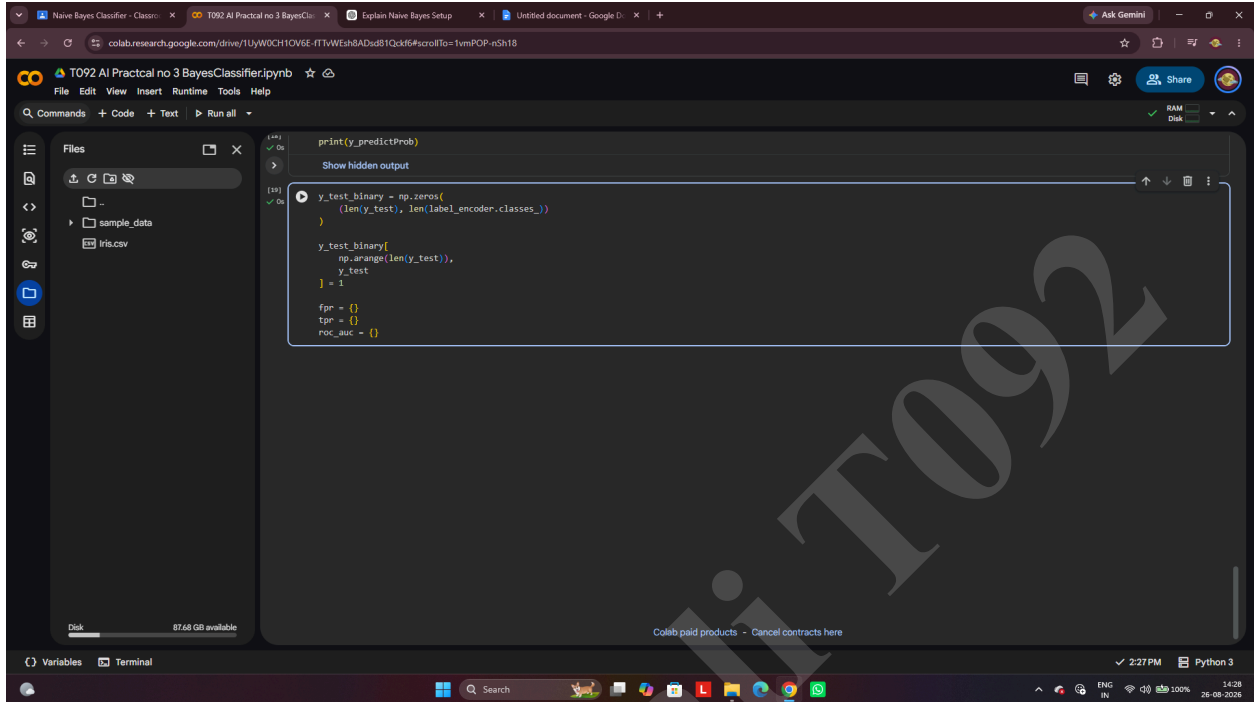
```
[[6.27617840e-035 9.99999877e-001 1.23101122e-007]
 [8.55355016e-079 9.99911933e-001 6.80473000e-005]
 [3.05150409e-204 2.11561867e-004 9.99788438e-001]
 [1.00000000e+000 2.05020488e-021 1.45997881e-026]
 [3.01347277e-018 9.99999918e-001 9.02504780e-001]
 [1.00000000e+000 5.97024055e-020 5.84774793e-025]
 [2.98903120e-146 3.92725418e-001 6.07274582e-001]
 [1.42859318e-213 3.64564793e-000 9.99996354e-001]
 [1.12582654e-151 2.53814729e-001 7.44562489e-001]
 [1.00000000e+000 3.35890293e-022 7.44349406e-026]
 [1.00000000e+000 1.20797480e-012 4.94237423e-019]
 [3.07809232e-220 5.45373563e-007 9.99999455e-001]
 [1.00000000e+000 3.61598450e-014 9.69586830e-020]
 [1.35314222e-108 9.95310989e-001 4.60901051e-003]
 [8.09123420e-116 9.32608296e-001 6.73317030e-002]
 [2.68577006e-116 8.24209080e-001 1.75730032e-001]
 [9.00118319e-190 1.47353871e-003 9.98526461e-001]
 [5.96740753e-005 9.99990961e-001 9.03807120e-000]
 [3.16136300e-191 1.79400046e-004 9.99820502e-001]
 [7.13349676e-172 3.73904582e-002 9.62600542e-001]
 [1.00000000e+000 3.37462489e-016 2.84819144e-022]
 [1.31376474e-090 9.09597604e-001 4.02387320e-004]
 [2.16160377e-145 7.44918657e-001 2.55801343e-001]
 [2.30271392e-110 9.95244303e-001 4.75569660e-003]
 [2.47551922e-081 9.99924796e-001 4.52050807e-005]
 [1.00000000e+000 7.47768356e-020 2.00200000e-025]
 [1.00000000e+000 7.16357713e-012 6.06095385e-018]
 [2.55065459e-208 4.17088866e-000 9.99995825e-001]
 [1.00000000e+000 1.87401976e-020 2.006083632e-023]
 [2.76012004e-212 6.39472143e-007 9.99999361e-001]
 [1.15821521e-044 9.99999899e-001 1.00554136e-007]
 [1.00530634e-121 8.61650914e-001 1.30430830e-001]
 [4.04160208e-005 9.99533107e-001 4.609251500e-005]
 [1.00000000e+000 3.91489556e-018 2.21278210e-024]
 [1.00000000e+000 8.97851511e-021 1.23761740e-026]
 [2.99208644e-166 4.48159330e-002 9.55184057e-001]
 [4.48802345e-122 9.94043756e-001 5.05624393e-003]]
```

Name :- Vinod Manohar Mali

Roll No :- T092

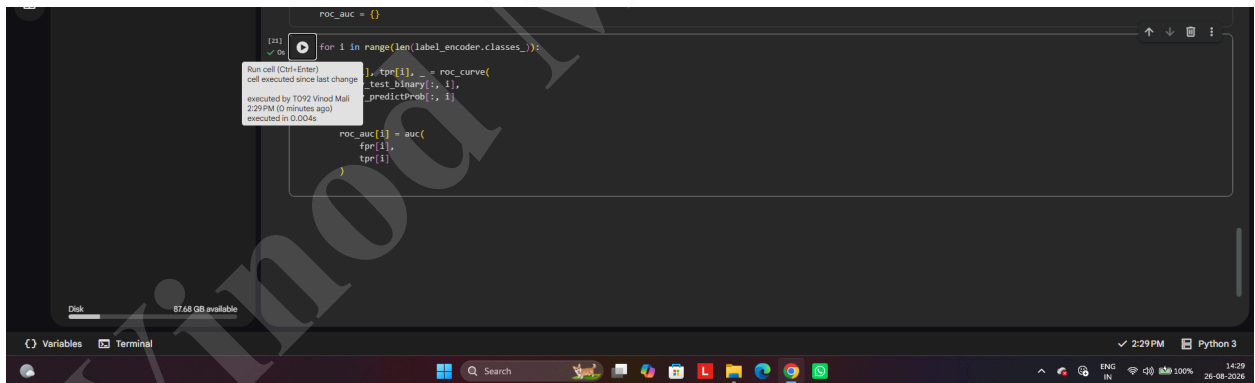
Sheth L.U.J & Sir M.V College Of Science
Subject :- Artificial Intelligences
Practical no 3

13. Prepare Data for ROC Curve



```
print(y_predictProb)
Show hidden output
y_test_binary = np.zeros(
    (len(y_test), len(label_encoder.classes_))
)
y_test_binary[
    np.arange(len(y_test)),
    y_test
] = 1
fpr = []
tpr = []
roc_auc = []
```

14. Calculate ROC and AUC



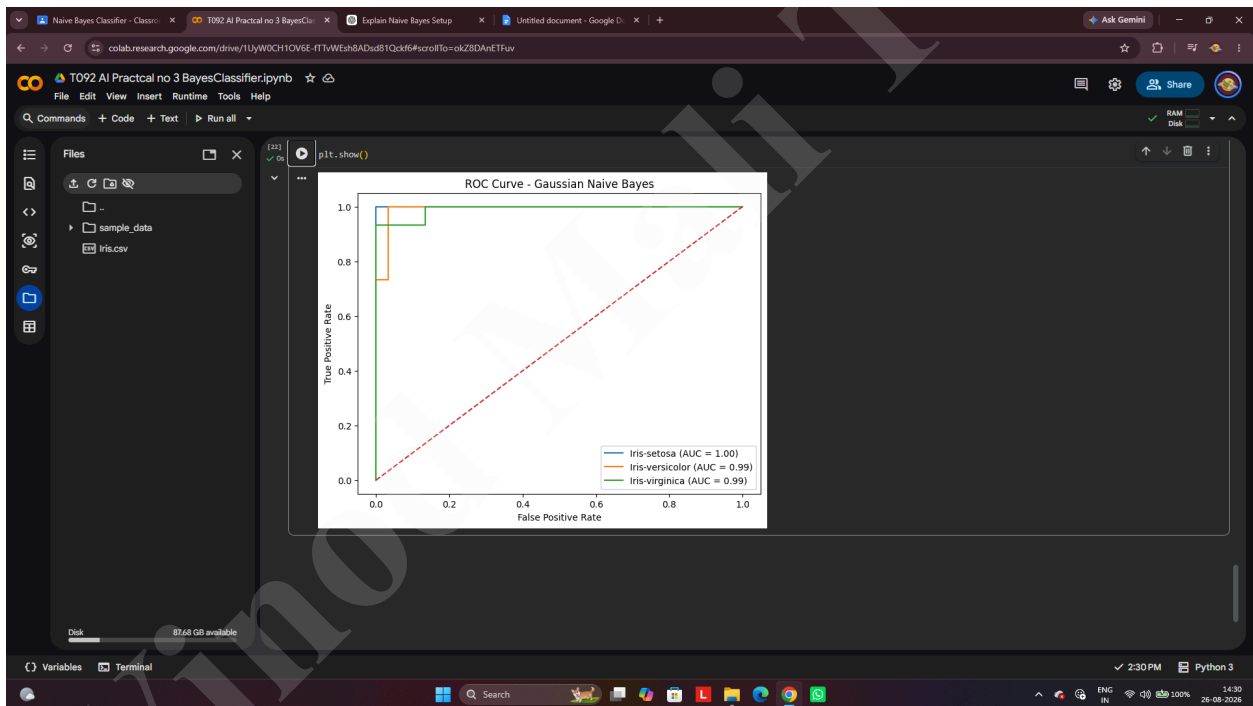
```
roc_auc = []
for i in range(len(label_encoder.classes_)):
    roc_auc[i] = auc(
        fpr[i],
        tpr[i]
    )
```

Name :- Vinod Manohar Mali
Roll No :- T092

Sheth L.U.J & Sir M.V College Of Science
Subject :- Artificial Intelligences
Practical no 3

15. Plot ROC Curve

```
fpr[i],  
tpr[i]  
)  
  
plt.figure(figsize=(8, 6))  
  
for i in range(len(label_encoder.classes_)):  
  
    plt.plot(  
        fpr[i],  
        tpr[i],  
        label='%s (AUC = %0.2f)' %  
            (label_encoder.classes_[i], roc_auc[i])  
    )  
  
    plt.plot(  
        [0, 1],  
        [0, 1],  
        linestyle='--'  
    )  
  
    plt.xlabel('False Positive Rate')  
    plt.ylabel('True Positive Rate')  
  
    plt.title('ROC Curve - Gaussian Naive Bayes')  
  
    plt.legend(loc='lower right')  
  
    plt.show()
```



Name :- Vinod Manohar Mali
Roll No :- T092